Chapter 3. High-Dimensional Data and Regularization

Predictive Modeling, Lecture 2 (OLS instability, Ridge, Lasso, Elastic Net, choosing λ)

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1 Linear Prediction and Why It Becomes Unstable

1.1 Setting

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad X \in \mathbb{R}^{n\times p}, \qquad \hat{y} = x_{\text{new}}^{\top}\hat{\boldsymbol{\beta}}.$$

Regularization picks, among $\hat{\boldsymbol{\beta}}$ that fit $\mathbf{y}$ well, one *less sensitive* on new x; stability is not causal identification.

1.2 High dimension is about p relative to n

Situation	Example	Problem for prediction
$p \ll n$	100,000 customers, 20 variables	fairly stable, but collinearity is still possible
p close to n	500 patients, 400 variables	small data changes move coefficients and selection a lot
$p \geq n$	1,000 documents, 50,000 words	OLS is not unique or cannot be computed
many derived features	raw variables, interactions, one-hot encodings	search degrees of freedom grow faster than signal

1.3 OLS and what it assumes

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 = (X^{\top}X)^{-1}X^{\top}\mathbf{y} \quad \text{(if } X^{\top}X \text{ is invertible)}.$$

It needs a well-conditioned $X^{\top}X$; otherwise this baseline *becomes unstable.*

1.4 When $p \geq n$: non-uniqueness

Theorem: interpolation is not unique

If $\text{rank}(X) < p$ (always the case when $p > n$), there is $v \neq 0$ with $Xv = 0$, and then $X(\hat{\boldsymbol{\beta}} + v) = X\hat{\boldsymbol{\beta}}$ for every solution $\hat{\boldsymbol{\beta}}$. In particular, when $\text{rank}X = n$, infinitely many $\boldsymbol{\beta}$ give $X\boldsymbol{\beta} = \mathbf{y}$ with zero training error.

$x_{\text{new}}^{\top}v$ need not be 0, so zero training error does not pick a good solution. (The minimum-norm interpolator $X^{\top}(XX^{\top})^{-1}\mathbf{y}$ is the $\lambda \to 0$ limit of Ridge.)

1.5 Collinearity and the variance of coefficients

Income, credit limit and card spending may measure one dimension: coefficients take opposite large signs and are reallocated across samples.

Theorem: variance along eigen-directions

Under $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I$, $\text{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2(X^{\top}X)^{-1}$. With the SVD $X = UDV^{\top}$, $D = \text{diag}(d_1, \ldots, d_p)$,

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid X) = \sigma^2 \sum_{j=1}^{p} \frac{v_j v_j^{\top}}{d_j^2}, \qquad \text{Var}(v_j^{\top}\hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{d_j^2}.$$

Proof [Supplement]. $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon} \Rightarrow \mathrm{Var} = \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1} = \sigma^2 V D^{-2} V^\top$. ■

$d_j \approx 0$ makes the variance along v_j explode, and $\Delta\hat{\boldsymbol{\beta}}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top \Delta\mathbf{y}$ amplifies perturbations. Check fold-wise stability of coefficients and selections, not only CV means.

2 Regularization: Buying Variance With Bias

$$\mathbb{E}[(Y - \hat{f}(X))^2] = \mathrm{Bias}^2 + \mathrm{Var} + \sigma^2.$$

In high dimensions OLS variance dominates; regularization trades bias for lower variance, judged by CV, not training loss. Too little tracks noise; too much shrinks the signal.

2.1 One template, three models

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \left[\frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda P(\boldsymbol{\beta}) \right], \qquad P(\boldsymbol{\beta}) = \begin{cases} \|\boldsymbol{\beta}\|_2^2 & \text{Ridge} \\ \|\boldsymbol{\beta}\|_1 & \text{Lasso} \\ \alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2 & \text{Elastic Net} \end{cases}$$

$\lambda = 0$ gives OLS; larger λ shrinks more. Choose λ by loss on new data.

2.2 Standardization comes first

$$x_{ij}^{\mathrm{std}} = \frac{x_{ij} - \mu_j^{(\mathrm{train})}}{s_j^{(\mathrm{train})}}.$$

Rescaling X_j by c rescales β_j by $1/c$, changing its penalty; standardization makes it unit-free. The intercept is not penalized (center first); one-hot and sparse inputs need judgement. μ_j, s_j come from the **training part only**.

3 Ridge

Motivation. Ridge shrinks most along poorly determined directions, trading a little bias for stability.

3.1 Closed form and constrained form

$$\hat{\boldsymbol{\beta}}_\lambda^{\mathrm{Ridge}} = \arg\min_{\boldsymbol{\beta}} \left[\frac{1}{n} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_2^2 \right] = (X^\top X + n\lambda I)^{-1} X^\top \mathbf{y}.$$

Proof [Supplement]. Gradient $-\frac{2}{n} X^\top (\mathbf{y} - X\boldsymbol{\beta}) + 2\lambda\boldsymbol{\beta} = 0 \iff (X^\top X + n\lambda I)\boldsymbol{\beta} = X^\top \mathbf{y}$; the matrix is positive definite for $\lambda > 0$, so the strictly convex objective has a unique minimizer even if $p \geq n$. ■

Coefficients shrink toward but not to 0; suited to spread-out signal. Equivalent (Lagrange) to $\min \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$ s.t. $\|\boldsymbol{\beta}\|_2^2 \leq c(\lambda)$.

3.2 Ridge shrinks small singular directions most

Theorem: SVD form of Ridge

With $X = UDV^\top$ and $\tilde{\lambda} = n\lambda$ (the slides absorb n into λ),

$$\hat{\boldsymbol{\beta}}^{\text{Ridge}} = V\,\operatorname{diag}\Big(\frac{d_j}{d_j^2+\tilde{\lambda}}\Big)U^\top \mathbf{y}, \qquad X\hat{\boldsymbol{\beta}}^{\text{Ridge}} = \sum_j u_j \frac{d_j^2}{d_j^2+\tilde{\lambda}} u_j^\top \mathbf{y}.$$

Proof [Supplement]. $X^\top X + \tilde{\lambda} I = V(D^2+\tilde{\lambda} I)V^\top$ and $X^\top \mathbf{y} = VDU^\top \mathbf{y}$, so $(X^\top X+\tilde{\lambda} I)^{-1}X^\top \mathbf{y} = V(D^2+\tilde{\lambda} I)^{-1}DU^\top \mathbf{y}$. ■

OLS multiplies direction j by $1/d_j$, Ridge by $d_j/(d_j^2+\tilde{\lambda}) \le 1/(2\sqrt{\tilde{\lambda}})$: small-$d_j$ directions shrink most. **Effective degrees of freedom** $\mathrm{df}(\lambda) = \sum_j d_j^2/(d_j^2+\tilde{\lambda})$ fall from p to 0.

3.3 Ridge shares signal among correlated variables

Theorem: identical columns get identical Ridge coefficients

If $x_{(1)} = x_{(2)}$, then $\hat{\beta}_1^{\text{Ridge}} = \hat{\beta}_2^{\text{Ridge}}$ for every $\lambda > 0$.

Proof [Supplement]. The fit depends only on $\beta_1+\beta_2$, so swapping $\beta_1 \leftrightarrow \beta_2$ preserves the strictly convex objective; its unique minimizer is symmetric. ■

OLS may give $\hat{y} \approx 12x_1 - 11x_2$; Ridge $\approx 0.8x_1 + 0.7x_2$ (illustrative): similar variables shrink together.

Data / goal	Why Ridge fits	What to check
many weak signals	keeps all coefficients a little	does it beat OLS?
strong multicollinearity	stabilizes unstable directions	standardization and coefficient paths
$p \ge n$ or sparse matrices	unique, computable solution	sparse implementation, memory
prediction over interpretation	targets generalization, not zeros	final metric and operating cost

4 Lasso

4.1 The L_1 penalty sets some coefficients exactly to zero

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\|\mathbf{y}-X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1\Big], \qquad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^p |\beta_j|.$$

Small coefficients become exactly 0 (**sparse** model); unselected does not mean irrelevant.

Theorem: soft-thresholding under an orthogonal design

If $X^\top X = nI$, let $z = X^\top \mathbf{y}/n$ (the OLS estimate). Then

$$\hat{\beta}_j^{\text{Lasso}} = S(z_j, \lambda/2) = \operatorname{sign}(z_j)(|z_j| - \lambda/2)_+, \qquad \hat{\beta}_j^{\text{Ridge}} = \frac{z_j}{1+\lambda}.$$

Proof [Supplement]. $\frac{1}{n}\|\mathbf{y}-X\boldsymbol{\beta}\|^2 = \frac{1}{n}\|\mathbf{y}\|^2 - 2\boldsymbol{\beta}^\top z + \|\boldsymbol{\beta}\|^2$, so the objective is $\sum_j\{(\beta_j - z_j)^2 + \lambda|\beta_j|\}$ + const. Subgradient: $2(\beta_j - z_j) + \lambda s_j = 0$, $s_j = \operatorname{sign}(\beta_j)$ if $\beta_j \neq 0$, $s_j \in [-1,1]$ if $\beta_j = 0$.

$|z_j| \le \lambda/2 \Rightarrow \beta_j = 0$; else $\beta_j = z_j - \frac{\lambda}{2}\text{sign}(z_j)$. Ridge: $(\beta_j - z_j)^2 + \lambda\beta_j^2$ is minimized at $z_j/(1+\lambda)$. ■

Ridge shrinks proportionally; Lasso subtracts a constant, zeroing small coefficients. Coordinate descent applies this soft-threshold to partial residuals.

4.2 Geometry: the ball and the diamond

Loss contours meet the smooth Ridge ball $\|\boldsymbol{\beta}\|_2^2 \le c$ off the axes, but the Lasso diamond $\|\boldsymbol{\beta}\|_1 \le c$ often at a corner, where some $\beta_j = 0$.

4.3 Lasso is unstable among correlated variables

Among variables measuring one factor, Lasso picks one, and a new sample may pick another. Check selection frequencies across folds; keep Ridge or Elastic Net as candidates.

Remark: a limit on Lasso selection

When $p > n$, the Lasso selects at most n variables. For two identical columns the Lasso solution is not unique: any split (β_1, β_2) with the same sum and signs is optimal.

Situation	Why Lasso is attractive	Caveat
variables costly to measure or deploy	simpler prediction pipeline	compare selection frequency and loss in performance
very many variables, sparse signal suspected	shrinks unnecessary coefficients to 0	may keep only one of correlated signals
an explainable operating model is needed	easy to present a variable list	do not turn selection into causal claims
screening for later analysis	starting point of exploratory reduction	needs separate validation and domain review

5 Elastic Net

$$\hat{\boldsymbol{\beta}}^{\text{EN}} = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\{\alpha\|\boldsymbol{\beta}\|_1 + (1-\alpha)\|\boldsymbol{\beta}\|_2^2\}\Big], \qquad \alpha \in [0,1].$$

$\alpha = 1$: Lasso; $\alpha = 0$: Ridge. The L_2 part gives strict convexity (unique solution) and a **grouping effect** (Exercise 3.3); the L_1 part keeps sparsity. Orthogonal design: $S(z_j, \lambda\alpha/2)/\{1 + \lambda(1-\alpha)\}$. Tune λ, α by CV.

6 Choosing λ Inside the Folds

Standardization and fitting per λ use only each fold's training part. Two rules:

- **minimum rule**: $\lambda_{\min}$ with the smallest mean CV loss;
- **1-SE rule**: $\lambda_{1\text{SE}} = \max\{\lambda : \widehat{R}_{\text{CV}}(\lambda) \le \widehat{R}_{\min} + \text{SE}_{\min}\}$, the most regularized model within one SE.

Fix the rule *before* the analysis; differences within CV noise are not meaningful.

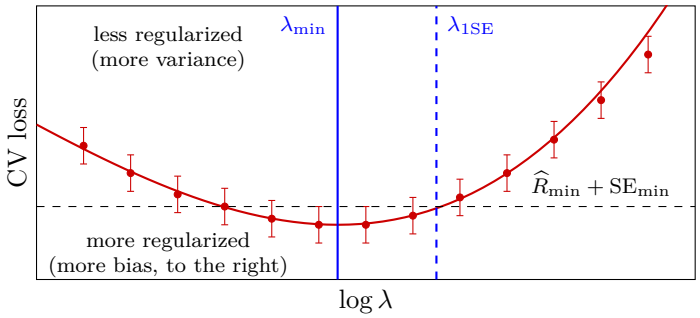


Figure 1: Schematic CV curve with ±1 SE bars: $\lambda_{1\mathrm{SE}}$ is the largest λ below the dashed line.

7 Regularized Logistic Regression

$$\hat{\boldsymbol{\beta}}_\lambda = \arg\min_{\boldsymbol{\beta}} \Big[\frac{1}{n}\sum_{i=1}^{n} \ell\{y_i, \sigma(\beta_0 + x_i^\top \boldsymbol{\beta})\} + \lambda P(\boldsymbol{\beta})\Big], \qquad \ell(y,p) = -[y\log p + (1-y)\log(1-p)].$$

Only the loss changes. With $\lambda > 0$ the objective is coercive, so a finite minimizer exists even under separation (Ch. 1).

8 Regularization Versus Robust Regression

Regularization	Robust regression
reduces coefficient variance and overfitting caused by many variables, collinearity and large degrees of freedom	mitigates loss domination caused by a few large residuals, label errors and extreme cases
Ridge, Lasso, Elastic Net	Huber, LAD
shrinkage, selection, generalization risk	influence of outliers, conditional location

Different failure causes; combinable (Huber loss + Elastic Net penalty).

9 Practical Selection and Reporting

Within one SE, prefer the simpler, more stable, faster candidate; one Lasso fit does not settle importance. Then refit on all training data and evaluate the test set once.

Example: reading a results table (illustrative)

Candidate	CV primary metric	Complexity	Interpretation
OLS	0.842 ± 0.031	180 variables	baseline; unstable under collinearity
Ridge	0.801 ± 0.019	180 shrunk coefficients	better performance and stability
Lasso	0.808 ± 0.026	24 selected	small loss in performance for simplicity
Elastic Net	0.799 ± 0.020	47 selected	compromise for correlated blocks

(Loss: smaller is better.)

Example: customer churn case

Situation. 8,000 customers, 600 derived variables; strong correlations among usage, payment and inquiry variables; churn rate 8%. **Candidates.** Standardized logistic baseline, Ridge, Lasso, Elastic Net. Primary metrics PR-AUC and log loss; operating metric recall@budget. **Split.** Protect the test period by time order; stratified CV within the training period. **Report.** CV mean and variability, chosen λ, number of selected variables, probability calibration and test performance in one table.

Summary

1. Many variables and collinearity make *coefficient variance and generalization risk*, rather than training fit, the main problem of OLS.
2. Ridge shrinks, Lasso selects, Elastic Net compromises; the right choice depends on the signal structure and the operating goal.
3. Standardization and λ selection are done inside CV folds; regularization does not automatically fix leakage or outliers.

10 Selected exercises

Motivation. Ridge risk, the Lasso path start $\lambda_{\max}$, and the Elastic Net grouping effect.

Exercise 3.1 (Some Ridge penalty always beats OLS)

Let X be fixed with $X^\top X = nI$, $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbb{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I$ and $\boldsymbol{\beta} \neq \mathbf{0}$. With the Ridge estimate $\hat{\boldsymbol{\beta}}_\lambda = z/(1+\lambda)$, $z = X^\top \mathbf{y}/n$, of the orthogonal-design theorem:
(a) show that $\mathbb{E}\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 = \dfrac{\lambda^2\|\boldsymbol{\beta}\|^2 + p\sigma^2/n}{(1+\lambda)^2}$;
(b) show that this risk decreases on $[0, \lambda^\star]$ and increases afterwards, where $\lambda^\star = p\sigma^2/(n\|\boldsymbol{\beta}\|^2)$, and that its minimum is strictly smaller than the OLS risk $p\sigma^2/n$.

Solution [Supplement].

(a)

Step 1. $z = X^\top(X\boldsymbol{\beta} + \boldsymbol{\varepsilon})/n = \boldsymbol{\beta} + X^\top\boldsymbol{\varepsilon}/n$, so $\mathbb{E}z = \boldsymbol{\beta}$ and $\mathrm{Var}(z) = \sigma^2 X^\top X/n^2 = (\sigma^2/n)I$.

Step 2. $\mathbb{E}\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta} = -\frac{\lambda}{1+\lambda}\boldsymbol{\beta}$, $\mathrm{Var}(\hat{\boldsymbol{\beta}}_\lambda) = \frac{\sigma^2}{n(1+\lambda)^2} I$; MSE = squared bias + trace:

$$\mathbb{E}\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2 = \frac{\lambda^2\|\boldsymbol{\beta}\|^2}{(1+\lambda)^2} + \frac{p\sigma^2}{n(1+\lambda)^2}.$$

(b)

Step 3. Write $a = \|\boldsymbol{\beta}\|^2 > 0$, $c = p\sigma^2/n$ and $f(\lambda) = (\lambda^2 a + c)/(1+\lambda)^2$. Then

$$f'(\lambda) = \frac{2\lambda a(1+\lambda) - 2(\lambda^2 a + c)}{(1+\lambda)^3} = \frac{2(\lambda a - c)}{(1+\lambda)^3},$$

negative for $\lambda < c/a = \lambda^\star$ and positive for $\lambda > \lambda^\star$.

Step 4. Substituting $\lambda^\star = c/a$,

$$f(\lambda^\star) = \frac{c^2/a + c}{(1 + c/a)^2} = \frac{ac}{a+c} < c = f(0).$$

The relative gain $c/(a+c)$ is largest at low signal-to-noise a/c. ■

Exercise 3.2 (When the Lasso selects nothing)

For a general design X and $\lambda > 0$, consider the Lasso objective $Q(\boldsymbol{\beta}) = \frac{1}{n}\|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 + \lambda\|\boldsymbol{\beta}\|_1$.
(a) Show that $\mathbf{0}$ minimizes Q if and only if $\lambda \geq \lambda_{\max} := \frac{2}{n}\|X^\top \mathbf{y}\|_\infty$.
(b) Show that all minimizers of Q have the same fitted values $X\hat{\boldsymbol{\beta}}$, and deduce that for $\lambda \geq \lambda_{\max}$ the minimizer $\mathbf{0}$ is unique.

Solution [Supplement].

(a)

Step 1. Q is convex, so $\boldsymbol{\beta}$ is a minimizer iff $\mathbf{0} \in \partial Q(\boldsymbol{\beta}) = -\frac{2}{n}X^\top(\mathbf{y} - X\boldsymbol{\beta}) + \lambda\,\partial\|\boldsymbol{\beta}\|_1$.

Step 2. $\partial\|\mathbf{0}\|_1 = [-1,1]^p$: $\frac{2}{n}X^\top\mathbf{y} \in \lambda[-1,1]^p \iff \frac{2}{n}|x_{(j)}^\top \mathbf{y}| \leq \lambda\ \forall j \iff \lambda \geq \lambda_{\max}$.

(b)

Step 3. If minimizers have $X\hat{\boldsymbol{\beta}}_1 \neq X\hat{\boldsymbol{\beta}}_2$, put $\bar{\boldsymbol{\beta}} = \frac{1}{2}(\hat{\boldsymbol{\beta}}_1 + \hat{\boldsymbol{\beta}}_2)$. Strict convexity of $u \mapsto \|\mathbf{y} - u\|^2$ and $\|\bar{\boldsymbol{\beta}}\|_1 \leq \frac{1}{2}(\|\hat{\boldsymbol{\beta}}_1\|_1 + \|\hat{\boldsymbol{\beta}}_2\|_1)$ give $Q(\bar{\boldsymbol{\beta}}) < \min Q$, a contradiction.

Step 4. For $\lambda \geq \lambda_{\max}$, $\mathbf{0}$ minimizes, so every minimizer has $X\hat{\boldsymbol{\beta}} = \mathbf{0}$ and $\frac{1}{n}\|\mathbf{y}\|^2 + \lambda\|\hat{\boldsymbol{\beta}}\|_1 = \min Q = \frac{1}{n}\|\mathbf{y}\|^2$, i.e. $\hat{\boldsymbol{\beta}} = \mathbf{0}$.

Hence the λ grid starts at $\lambda_{\max}$ (orthogonal design: $|z_j| \leq \lambda/2$ for all j). ■

Exercise 3.3 (Grouping effect of the Elastic Net)

Let $\mathbf{y}$ be centered and the columns of X standardized ($\mathbf{1}^\top x_{(j)} = 0$, $\|x_{(j)}\|^2 = n$), with sample correlation $\rho_{ij} = x_{(i)}^\top x_{(j)}/n$. Let $0 \leq \alpha < 1$, $\lambda > 0$, and $\hat{\boldsymbol{\beta}}$ the Elastic Net estimate. Show that if $\hat{\beta}_i\hat{\beta}_j > 0$, then

$$|\hat{\beta}_i - \hat{\beta}_j| \leq \frac{\|\mathbf{y}\|}{\sqrt{n}} \cdot \frac{\sqrt{2(1-\rho_{ij})}}{\lambda(1-\alpha)}.$$

Solution [Supplement].

Step 1. With $\hat{r} = \mathbf{y} - X\hat{\boldsymbol{\beta}}$, for $\hat{\beta}_k \neq 0$ the objective is differentiable in β_k:

$$-\frac{2}{n}x_{(k)}^\top \hat{r} + \lambda\alpha\,\mathrm{sign}(\hat{\beta}_k) + 2\lambda(1-\alpha)\hat{\beta}_k = 0.$$

Step 2. Equal signs for $k = i, j$; subtract:

$$2\lambda(1-\alpha)(\hat{\beta}_i - \hat{\beta}_j) = \frac{2}{n}(x_{(i)} - x_{(j)})^\top \hat{r}.$$

Step 3. By Cauchy–Schwarz, $|\hat{\beta}_i - \hat{\beta}_j| \leq \|x_{(i)} - x_{(j)}\|\,\|\hat{r}\|/\{n\lambda(1-\alpha)\}$, and $\|x_{(i)} - x_{(j)}\|^2 = 2n - 2n\rho_{ij}$.

Step 4. Comparing with $\mathbf{0}$: $\frac{1}{n}\|\hat{r}\|^2 \leq \frac{1}{n}\|\hat{r}\|^2 + \lambda P(\hat{\boldsymbol{\beta}}) \leq \frac{1}{n}\|\mathbf{y}\|^2$, so $\|\hat{r}\| \leq \|\mathbf{y}\|$ and

$$|\hat{\beta}_i - \hat{\beta}_j| \leq \frac{\sqrt{2n(1-\rho_{ij})}\,\|\mathbf{y}\|}{n\lambda(1-\alpha)} = \frac{\|\mathbf{y}\|}{\sqrt{n}} \cdot \frac{\sqrt{2(1-\rho_{ij})}}{\lambda(1-\alpha)}.$$

For the Lasso ($\alpha = 1$) the bound is void. ■